

VisLinx-M User Manual

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1 Revision History

Ver	Date	Notes
1.0	04.2024	Initial release
1.1	01.2025	Metadata Information section was supplemented
1.2	03.2025	Metadata Information section supplemented. Image Format Control section updated. Output buffer section added
2.1	04.2025	Updated Image Format Control section - added TDI Y Shift parameter.
2.2	05.2025	Added "Connection requirement" section
2.3	11.2025	Updated Image Format Control section

Table 1 – Revision History

Table of Contents

1	REVISION HISTORY	1
2	FIGURES & TABLES	3
2.1	List of Figures	3
2.2	List of Tables	3
3	INTRODUCTION.....	4
3.1	Safety Precautions	4
3.2	Disclaimer	5
4	OVERVIEW	6
5	CONNECTION REQUIREMENT	7
5.1	PCIe device connection.....	7
5.2	Thunderbolt connection	8
6	CONFIGURATION INTERFACE.....	9
6.1	Device Control	9
6.2	Image Format Control.....	10
6.2.1	Sensor region sampling geometry	13
6.2.2	TDI Mode	14
6.3	Acquisition Control	15
6.3.1	Exposure Time	16
6.4	Analog Control	17
6.4.1	Analog and Digital Gain and Black Level	17
6.5	Pixel Correction Control.....	18
6.5.1	Defect Pixel Correction	19
6.6	Test Control	20
6.6.1	Build-In-Test.....	21
6.7	Metadata Information	22
6.8	Output Buffer.....	24
6.8.1	Visual buffer representation.....	24
6.8.2	Buffer size calculations	25
6.9	I2C bus interface	26
	REFERENCES.....	27

2 Figures & Tables

2.1 List of Figures

Figure 1 – VisLinx-M connection using VislinxM PCIe adapter	7
Figure 2 – VisLinx-M connection using VislinxM PCIe adapter and HLCD cable	7
Figure 3 – Example of Physical Isolation of 7 Pins to Adjust PCIe Adapter to x2 Lanes	7
Figure 4 – Sensor regions.....	13
Figure 5 – TDI operation	14
Figure 6 – Defect pixel correction position for Mono image	19
Figure 7 – Defect pixel correction position for Color image	19
Figure 8 – Raw data frame structure	23
Figure 9 – Partial band selection in full frame representation	24
Figure 10 – Full frame representation with all bands enabled	24

2.2 List of Tables

Table 1 – Revision History.....	1
Table 2 – Device Control parameters	9
Table 3 – Image Format control parameters	11
Table 4 – Acquisition Control parameters	15
Table 5 – Analog Control parameters	17
Table 6 – Pixel Correction Control parameters.....	18
Table 7 – Test Control parameters	20
Table 8 – ChunkDataControl Gen<i>Cam parameters	22
Table 9 – Frame MetaData structure.....	23

3 Introduction

3.1 Safety Precautions

Please take the time to read through the precautions listed below to avoid preventable and unnecessary injuries and damage to you, other personnel, or property. Read these safety instructions carefully before your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual to prevent misuse of the product.



Caution! Read Carefully and do not disregard these instructions.

In the event of a failure, disconnect the power supply

Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.

If an unpleasant smell or smoking occurs, disconnect the power supply.

Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.

Do not disassemble, repair or modify the product.

Such actions may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel before inspection, modification, or repair.

Do not place the product on unstable surfaces.

Otherwise, it may drop or fall, resulting in injury to persons or the camera.

Do not use the product if dropped or damaged.

Otherwise, a fire or electric shock may occur.

Do not touch the product with metallic objects.

Otherwise, a fire or electric shock may occur.

Do not place the product in dusty or humid environments, nor where water may splash.

Otherwise, a fire or electric shock may occur.

Do not wet the product or touch it with wet hands.

Otherwise, the product may fail or cause a fire, smoking, or electric shock.

Do not touch the gold-plated sections of the connectors on the product.

Otherwise, the surface of the connector may be contaminated by sweat or skin oil, resulting in contact failure of a connector, malfunction, fire, or electric shock due to static electricity discharge.

Do not use or place the product in the following locations.

- Unventilated areas such as closets or bookshelves
- Near oils, smoke, or steam
- Next to heat sources
- A car with closed doors where the temperature can become hot
- Static electricity replete locations
- Near water or chemicals

Otherwise, a fire, electric shock, accident, or deformation may occur due to a short circuit or heat generation.

Do not place heavy objects on the product.

Otherwise, the product may be damaged.

Be sure to discharge static electricity from the body before touching any sensitive electronic components.

The electronic circuits in your computer and the circuits on the *Iron* camera and the *Predator II* board are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

3.2 Disclaimer

This product should only be used for image capturing and processing. **KAYA Instruments** assumes no responsibility for any damage that may ensue by using the camera for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company takes no responsibility for any damages caused by:

- Earthquake, thunder strike, natural disasters, a fire caused by usage beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions
- Secondary damages caused by the use of this product or its unusable state (business interruption or others)
- Use of this product in any manner that contradicts this manual or malfunctions due to connection to other devices.
- Damage to this product that is out of our control or failure due to modification.
- Accidents and/or third parties that may be involved.

Additionally, **KAYA Instruments** assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product
- Any consequences or other abnormalities following the use of this product

Repairs to this product are carried out by replacing it on a chargeable basis and not by repairing the faulty device. Non-chargeable replacement is offered for the initial failure, as long as it is reported no later than two weeks post-delivery of the product.

4 Overview

This document describes the functionality and features of the VisLinx-M camera. Connection and streaming can be achieved in a few easy steps and requires a few configurations. The cameras provide a vast variety of image processing algorithms and configurations to adjust the stream output. Camera control can be achieved using the standard Gen*i*Cam interface, subordinate to the camera's descriptive schema (XML) file.

5 Connection requirement

5.1 PCIe device connection

VisLinux-M PCIe device requires a connection via x2 lanes of PCIe Gen 3.0.

Please adjust the number of PCIe lanes used by the device either in the BIOS settings or by physically isolating the PCIe contact lines on the device's connection board.

It is crucial that the device is connected and operating strictly at PCIe 3.0 x2 lanes to ensure stable performance.

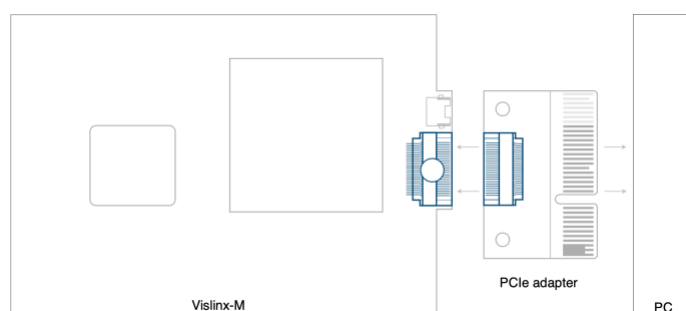


Figure 1 – VisLinux-M connection using VislinxM PCIe adapter

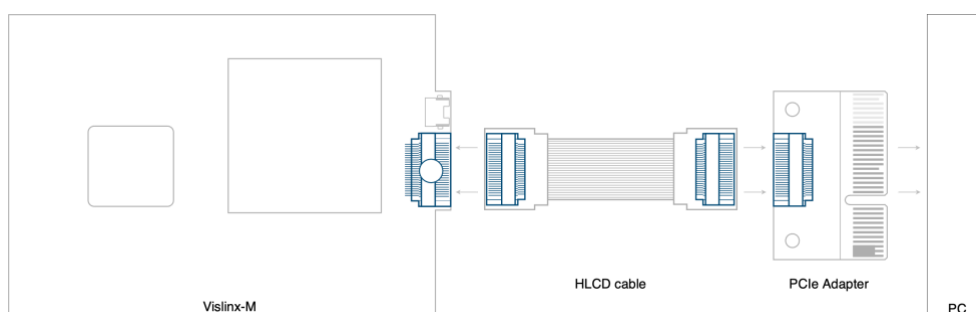


Figure 2 – VisLinux-M connection using VislinxM PCIe adapter and HLCD cable

To configure a PCIe adapter to operate in x2 lane mode, physically isolate specific PCIe lane pins on the edge connector. This method is typically used when BIOS configuration is not possible or insufficient.

Warning: Physical modification can void warranties and should be done only by professionals.



Figure 3 – Example of physical isolation of 10 pins to adjust PCIe adapter to x2 Lanes

5.2 Thunderbolt connection

Thunderbolt is an interface that enables PCIe devices to be powered and operated over a cable. Please note that using this interface should be treated like standard PCIe device:

- It does not support hot-swapping and must be connected before system initialization (i.e., before power-up and POST).
- It may not be compatible with hibernation or sleep modes, which is why we strongly recommend [disabling Fast Startup on Windows OS](#) and disabling Fast Boot is a feature in BIOS.
- When using an external Thunderbolt enclosure, make sure to perform an extra power cycle on the enclosure itself because it may be not dependent on the host system.
- Please ensure that the PCIe connection is stable before conducting any tests. Connection instability can be caused by extended Thunderbolt or PCIe riser cable length.

6 Configuration Interface

6.1 Device Control

The Device Control contains manufacturer parameters describing the currently connected hardware device. The information includes the device vendor name, basic manufacturer information details, and the currently running firmware version. This information can be used to identify the specific hardware and notify in case a firmware update is needed to support the complete functionality set.

Parameter	Description	Gen <i><i>Cam</i> name	Type	Possible values		Remarks
				Value	Gen <i><i>Cam</i> name	
Gen <i><i>Cam</i> Category: DeviceControl						
Device Vendor Name	Name of the manufacturer of the device	DeviceVendorName	String			
Device Model Name	The model of the device	DeviceModelName	String			
Device Manufacturer Info	Extended manufacturer information about the device	DeviceManufacturerInfo	String			
Device Version	The version of the device	DeviceVersion	String			
Device Firmware Version	The firmware version of the device	DeviceFirmwareVersion	String			
Device Serial Number	Device's serial number. This string is a unique identifier of the device	DeviceSerialNumber	String			
Device Temperature Selector	Selects the temperature value source	DeviceTemperatureSelector	Enumeration (Selector)	0 1	Processor Sensor	
Device Temperature	Device temperature	DeviceTemperature [DeviceTemperatureSelector]	Float	Max: 120 Min: -60		In °C

Table 2 – Device Control parameters

6.2 Image Format Control

The Image Format Control is responsible for defining the output image dimensions and format type. The resolution of the image and output format will influence the maximum frame rate which can be achieved.

Parameter	Description	Gen <i><i>Cam</i> name	Type	Possible values		Remarks
				Value	Gen <i><i>Cam</i> name	
Gen <i><i>Cam</i> Category: ImageFormatControl						
Width Min	Minimum width of the image	WidthMin	Integer			In pixels
Height Min	Minimum height of the image	HeightMin	Integer			In lines
Width Max	Maximum width of the image	WidthMax	Integer			In pixels
Height Max	Maximum height of the image	HeightMax	Integer			In lines
Width	Width of the image provided by the device (in pixels)	Width	Integer			
Height	Height of the image provided by the device (in pixels)	Height	Integer			Not used
Pixel Format	Indicates the format of the pixel to use during the acquisition	PixelFormat	Enumeration	0x01100003	Mono10	
Test Pattern	Selects the type of test pattern that is generated by the device as image source	TestPattern	Enumeration	0x00 0x01	Off SensorGrayDiagonalRamp	
Scan Type	Scan type of the sensor of the device	DeviceScanType	Enumeration	0	Areascan	
Region 0 Mode	Controls if the Region 0 of interest is active and streaming	Region0Mode	Enumeration	0 1	Off On	
Region 1 Mode	Controls if the Region 1 of interest is active and streaming	Region1Mode	Enumeration	0 1	Off On	
Region 2 Mode	Controls if the Region 2 of interest is active and streaming	Region2Mode	Enumeration	0 1	Off On	
Region 3 Mode	Controls if the Region 3 of interest is active and streaming	Region3Mode	Enumeration	0 1	Off On	
Region 4 Mode	Controls if the Region 4 of interest is active and streaming	Region4Mode	Enumeration	0 1	Off On	
Region 5 Mode	Controls if the Region 5 of interest is active and streaming	Region5Mode	Enumeration	0 1	Off On	
Region 6 Mode	Controls if the Region 6 of interest is active and streaming	Region6Mode	Enumeration	0 1	Off On	
Region 0 OffsetY	Region 0 vertical offset from the origin to the region of interest	Region0OffsetY	Integer	Min: 0 Max: HeightMax-1		In lines <i>See remark (4)</i>
Region 1 OffsetY	Region 1 vertical offset from the origin to the region of interest	Region1OffsetY	Integer	Min: 0 Max: HeightMax-1		In lines <i>See remark (4)</i>

Region 2 OffsetY	Region 2 vertical offset from the origin to the region of interest	Region2OffsetY	Integer	Min: 0 Max: HeightMax-1	In lines <i>See remark (4)</i>
Region 3 OffsetY	Region 3 vertical offset from the origin to the region of interest	Region3OffsetY	Integer	Min: 0 Max: HeightMax-1	In lines <i>See remark (4)</i>
Region 4 OffsetY	Region 4 vertical offset from the origin to the region of interest	Region4OffsetY	Integer	Min: 0 Max: HeightMax-1	In lines <i>See remark (4)</i>
Region 5 OffsetY	Region 5 vertical offset from the origin to the region of interest	Region5OffsetY	Integer	Min: 0 Max: HeightMax-1	In lines <i>See remark (4)</i>
Region 6 OffsetY	Region 6 vertical offset from the origin to the region of interest	Region6OffsetY	Integer	Min: 0 Max: HeightMax-1	In lines <i>See remark (4)</i>
Region Height	Height of the image provided by the device (in pixels) in multi-roi mode	RegionHeight	Integer	Min: 0 Max: HeightMax-1	In lines
TDI Stages	Number of TDI stages to be accumulated in TDI mode	TDIStages	Integer	Min: 1 Max: 32	<i>See remark (1) and (2)</i>
TDI Mode	Mode of TDI operation	TDIMode	Enumeration	0 1 2	Off Record TDI
TDI Maximum Frame Count	Number of maximum frames that can be acquired in Record TDI mode	TDIFrameCountMax	Integer		
TDI Frame Count	Number of frames to acquire in Record TDI mode	TDIFrameCount	Integer	Min: 1 Max: TDIFrameCountMax	<i>See remark (3)</i>
Band X Shift	Satellite movement horizontal shift	BandXShift	Integer	Min: -64 Max: 64	In pixels
Band Y Shift	Satellite movement vertical shift	BandYShift	Integer	0	In lines. Currently not used.
TDI Y Shift	Shift Y of TDI stages for merge	TDIYShift	Integer		In lines <i>See remark (5)</i>
Horizontal Invert	Flip horizontally the image sent by the device	ReverseX	Boolean	0 – false 1 - true	
Vertical Invert	Flip vertically the image sent by the device	ReverseY	Boolean	0 – false 1 - true	

Table 3 – Image Format control parameters

Remarks:

The following restrictions must be applied; otherwise, this may cause unexpected frame rate, corrupted image, overlapping data between the bands etc.

- "TDIStages" is only available in TDI mode i.e "TDIMode" == "TDI" (2)
- When operating at low TDI stages the frame rate will be reduced
- "TDIFrameCount" is only available in Record mode i.e "TDIMode" == "Record" (1)
- "Region0OffsetY", "Region1OffsetY",... etc.
 - Must be in ascending order otherwise it may cause unexpected results.
 - Value must be between [64] and [7000 - 64 - "RegionHeight"] in order to provide functionality of the BandYShift.

- Difference between two adjacent offsets must be at least $(128 + \text{"RegionHeight"})$ in order to provide functionality of the *BandYShift*.
- 5. When *"TDIStages"* or *"RegionHeight"* are changed, the *"TDIYShift"* value is updated automatically to *"RegionHeight"* divided by *"TDIStages"*; then it is possible to update this parameter's value.

6.2.1 Sensor region sampling geometry

The geometry of the sensor and regions is illustrated in the figure below:

1. All bands have the same width and region height.
2. OffsetX of all regions is the same and calculated in such way that the ROI is shifted to the middle of the sensor.
3. Active bands are stacked in the buffer and output to the stream.
4. The metadata line is added at the end of the buffer as an additional line.
5. The “Height” parameter represents the total buffer height, including the additional metadata line.

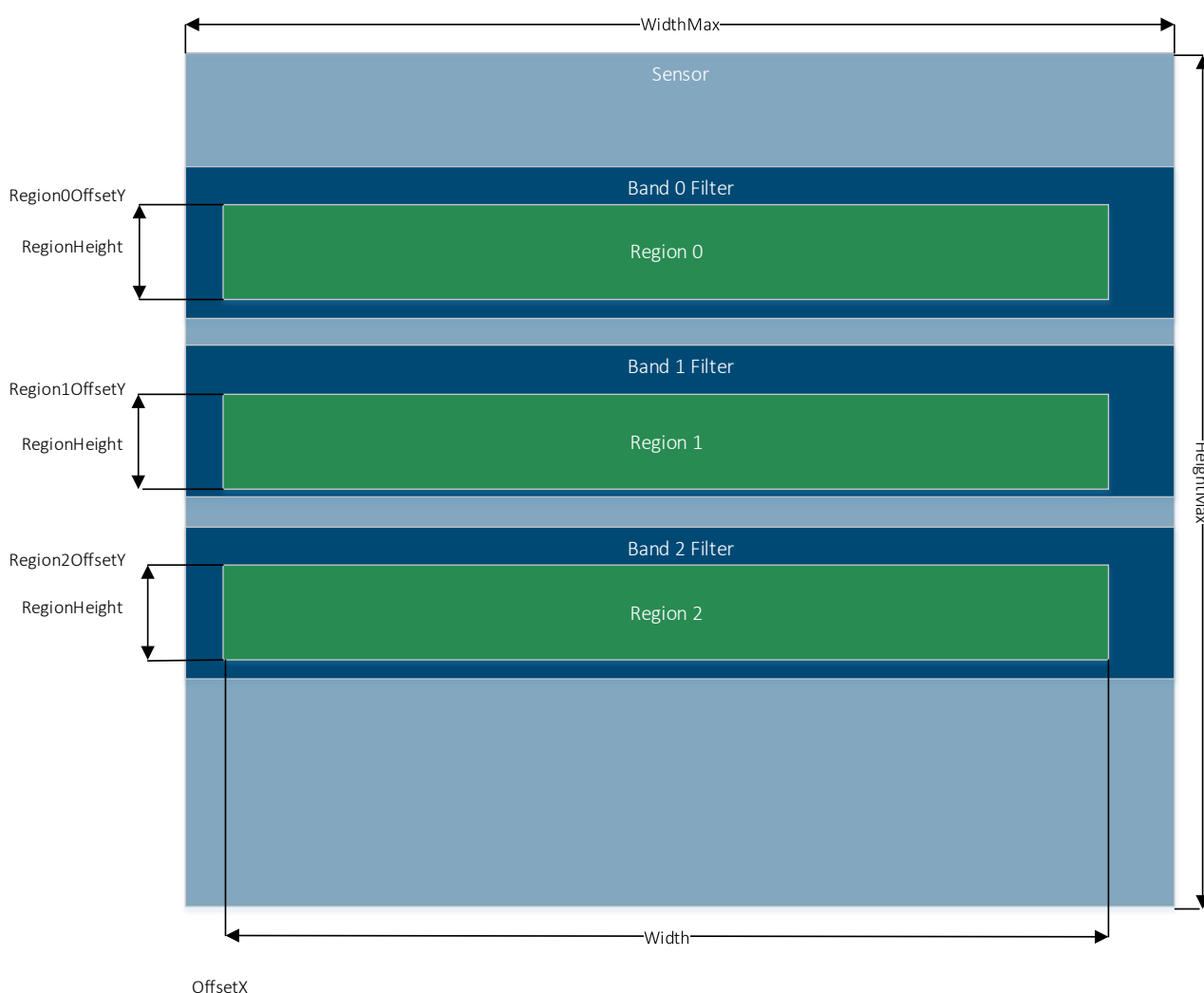


Figure 4 – Sensor regions

6.2.2 TDI Mode

Defines the TDI operation mode according to the following:

1. Off (Video):
Data is streamed through DRAM without any changes . Frame rate is limited
2. Record:
Data is recorded in DRAM without any changes up to number of specified frames (). Then it is readout.
3. TDI:
A digital TDI operation
 - Each band frame is written to DRAM with overlap with the previous frame.
 - The shift between overlaps is TDIShift which is Region Height divided by TDI Stages.
 - The previous non overlapping region from frame N-1 is readout to PCIe bus.

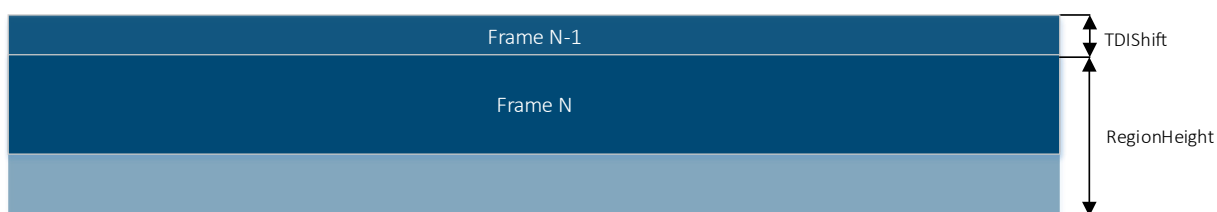


Figure 5 – TDI operation

6.3 Acquisition Control

The Acquisition stream control section describes settings and states for data generation (commands and stream). The acquisition can also be controlled through executing the relevant commands from this category.

Parameter	Description	Gen <i><i>Cam name</i>	Type	Possible values		Remarks
				Value	Gen <i><i>Cam name</i>	
Gen <i><i>Cam Category: AcquisitionControl</i>						
Acquisition Reset	Resets the image acquisition logic and counters	AcquisitionReset	Command	1 - Activate		
Acquisition Start	Starts the Acquisition of the device	AcquisitionStart	Command	1 - Activate		
Acquisition Stop	Stops the Acquisition of the device at the end of the current Frame	AcquisitionStop	Command	0 - Activate		
Frame Rate Max	Acquisition rate maximum value	AcquisitionFrameRateMax	Float			In units of Hz
Frame Rate	Controls the acquisition rate at which the frames are captured	AcquisitionFrameRate	Float			In units of Hz
Exposure Mode	Sets the operation mode of the Exposure (or shutter)	ExposureMode	Enumeration	0x00	Timed	Not used
Exposure Time Max	Exposure time maximum value	ExposureTimeMax	Float			In units of microseconds (us)
Exposure Time	Sets the Exposure time when ExposureMode is Timed	ExposureTime	Float			In units of microseconds (us) <i>See remark (1)</i>
Trigger Selector	Selects the type of trigger to configure	TriggerSelector	Enumeration	0	FrameStart	
Trigger Mode	Controls if the selected trigger is active	TriggerMode	Enumeration	0x00 0x01	Off On	
Trigger Source	Specifies the internal signal or physical input Line to use as the trigger source	TriggerSource	Enumeration	0x01	Software	

Table 4 – Acquisition Control parameters

Remarks:

1. Controls the duration where the photosensitive cells are exposed to light.

6.3.1 Exposure Time

Time in microseconds [μsec] in which sensor is exposed to light. This time is subject to the specified image frame rate:

Minimum exposure depends on camera model, connection configuration and sensor characteristics

$$\text{Maximum exposure} \cong \frac{1,000,000}{\text{frame rate}}$$

Additional delays might be taken into consideration in the calculation of exposure values, like the delay between frames, etc.

6.4 Analog Control

Analog control parameters describes how to influence the analog sensor features and digital modifiers, such as gain, black level, white balance and voltages, to manipulate image output.

6.4.1 Analog and Digital Gain and Black Level

Parameter	Description	Gen <i><i></i> Cam name	Type	Possible values		Remarks
				Value	Gen <i><i></i> Cam name	
Gen <i><i></i> Cam Category: AnalogControl						
Gain Selector	Selects which Gain is controlled by the various Gain features	GainSelector	Enumeration 0 (Selector)	DigitalAll		
Gain	Controls the selected gain as an absolute physical value	Gain [GainSelector]	Float	Max. 7.99923		
Analog Gain Level	Controls the analog gaining level	AnalogGainLevel	Float			See remark (1)
Black Level Selector	Selects which Black Level is controlled by the various Black Level features	BlackLevelSelector	Enumeration 0	All		
Black Level	Controls the digital black level as an absolute physical value	BlackLevel [BlackLevelSelector]	Enumeration	Min. -4096 Max. 4095		Depends on selected pixel format
Black Level Auto	Controls the mode for automatic black level adjustment	BlackLevelAuto	Enumeration	0x00 0x01	Off AnalogContinuous	
Analog Black Level	Controls the analog black level as an absolute physical value. Represents the applied DC offset	AnalogBlackLevel	Float	Min. -8192 Max. 8191		See remark (1)

Table 5 – Analog Control parameters

Remarks:

1. The values of the Analog Gain and Analog Black Level parameters depends on sensor's analog range

The Analog Control parameters can be used to control and adjust the gain and the black level available features.

Where P is the pixel that is being corrected, the Black Level is the offset of said pixel and the Gain is the gain of the pixel. Analog Black Level Controls the analog black level as an absolute physical value. Represents the applied DC offset.

6.5 Pixel Correction Control

The pixel correction control allows compensating any sensor dead pixel by averaging adjacent pixels. The Dark and Flat field correction algorithm helps to solve issues with fixed pattern noise, which usually originates from the sensor.

Parameter	Description	Gen <i><i>Cam name</i>	Type	Possible values		Remarks
				Value	Gen <i><i>Cam name</i>	
Gen <i><i>Cam Category: PixelCorrectionControl</i>						
Defect Pixel Correction Enable	Enable the Defect Pixel correction algorithm	DefectPixelCorrectionEnable	Boolean	0 - false 1 - true		
Defect Pixel Selector Max	Total number of defect pixels to be corrected	DefectPixelSelectorMax	Integer			
Defect Pixel Selector	Total number of defect pixels to be corrected	DefectPixelSelector	Integer (Selector)			
Defect pixel X coordinate	Configure defect pixel X coordinate	DefectPixelX [DefectPixelSelector]	Integer	Min: -1 Max: SensorWidth		
Defect pixel Y coordinate	Configure defect pixel Y coordinate	DefectPixelY [DefectPixelSelector]	Integer	Min: -1 Max: SensorHeight		
Defect Pixel Remove	Remove the defect pixel determined by DefectPixelWriteX and DefectPixelWriteY	DefectPixelRemove [DefectPixelSelector]	Command	1 - Activate		

Table 6 – Pixel Correction Control parameters

6.5.1 Defect Pixel Correction

The defected pixel correction will correct up to 32 pixels in the sensor and up to 2 adjacent pixels in a row. The pixel correction coordinates represent pixels of the sensor's visible ROI, therefore identifying the correct X and Y coordinate should be done using a default, full resolution image.

The algorithm will correct the defect pixel based on the value of existing adjacent pixels. The correction for the Mono and Color sensor is slightly different and described as follows:

Mono pixel correction:

The defect pixel $P(x,y)$ value will be the average value of 2 pixels adjacent to pixel $P(x,y)$ from both sides in the same row.

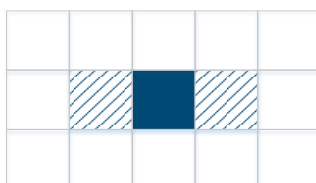


Figure 6 – Defect pixel correction position for Mono image

Color (Bayer) pixel correction:

The defect pixel $P(x,y)$ value will be the average value of two pixels from both sides of pixel $P(x,y)$ in the same row, corresponding to the same Bayer color element.

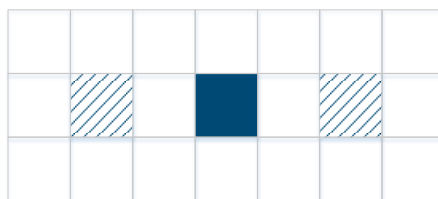


Figure 7 – Defect pixel correction position for Color image

6.6 Test Control

The test Control category contains parameters for camera testing and analysis purposes only. These configurations are not required for the standard operation of the camera. Configuration of these parameters might result in unexpected camera behavior if the wrong value is input.

Parameter	Description	Gen <i><i>Cam</i> name	Type	Possible values		Remarks
				Value	Gen <i><i>Cam</i> name	
Gen <i><i>Cam</i> Category: TestControl						
Sensor Register Address	Sensor Register Address	SensorRegAddress	Integer			
Sensor Register Value	Returns the value of sensor register at SensorRegAddress address	SensorRegValue	Integer			
Gen <i><i>Cam</i> Category: TestControl/BIT						
BIT Reset All	Reset all BITs	BITResetAll	Command	1 - Activate		
BIT Start All	Start all BITs	BITStartAll	Command	1 - Activate		
BIT Count	Number of available BITs	BITCount	Integer			
BIT Selector Index	Selects BIT configuration	BITSelectorIndex	Integer (Selector)			
BIT Selector	Selects BIT configuration	BITSelector	Enumeration	0	Flash	
				2	SensorControl	
				3	SensorLVDS	
				4	Temperature	
				5	Voltages	
BIT Start	Start selected BIT	BITStart	Command	1 - Activate		
BIT Status	BIT current status	BITStatus	Enumeration	0	Unknown	
				1	Pass	
				2	Fail	
				0xFF	Unsupported	
BIT Error Report	BIT last error report description	BITErrorReport	String			

Table 7 – Test Control parameters

6.6.1 Build-In-Test

The Build-In-Test (BIT) implements an option to check individual camera's interfaces to insure correct behavior in the allowed range. Each test may result in "Pass", "Fail" or "Unsupported" with an appropriate error report, either by starting an individual test or all tests together.

Following describe each test functionality:

1. Flash – Test the functionality of flash access.
2. SensorControl – Test basic communication with the sensor by reading and comparing it with a known default value.
3. SensorLVDS – Test stream interface with the sensor using a known pattern.
4. Temperature – Test the temperature of several components; they must be in the acceptable range.
5. Voltages – Test the analog voltages levels; they must be in the acceptable range.
6. DRAM – Test the external DRAM memory access and functionality.

6.7 Metadata Information

Runtime metadata information can be attached to each transmitted frame band. When enabled, the metadata will be added as a single line to the end of each frame band. The information will be relevant for the specific frame and will be updated in the next frame.

Parameter	Description	Gen <i><i>Cam</i> name	Type	Possible values		Remarks
				Value	Gen <i><i>Cam</i> name	
Gen <i><i>Cam</i> Category: ChunkDataControl						
Chunk Mode Active	Activates the inclusion of Chunk data in the payload of the image	ChunkModeActive	Boolean			
Chunk Satellite Bus	Satellite Bus Metadata	ChunkSatelliteBus	Register (48 bytes)			This one is used to get/set all the metadata at once
XDLX	XDlinx Labs Identity	ChunkXDLX	String			
SAT_ID	Satellite ID	ChunkSatID	String			
Orbit Number	Present Orbit Number	ChunkOrbitNumber	Integer			
Task ID	Unique Job/Task ID Generated by Ground	ChunkTaskID	Integer			
Image Start Time	Imaging Start time (Seconds level Accuracy)	ChunkImageStartTime	Integer			
Imaging Duration	Imaging duration (Seconds level Accuracy)	ChunkImagingDuration	Integer			
Config/TDI File	Camera Configuration/TDI ID	ChunkConfigTDIFile	Integer			Camera Configuration ID (max 16 ID); TDI Configurations (max 16 ID)
Latitude	Image Frame central Latitude (single precision Floating point)	ChunkLatitude	Float			
Longitude	Image Frame central Longitude (single precision Floating point)	ChunkLongitude	Float			
PPS Sync Ref	Seconds level precision for the duration of mission life. To be filled by Grabber	ChunkPPSRef	Integer			Read only . Set during I2C Set comand
Time Counter Ref	8 ns level precision for the duration of mission life. To be filled by Grabber	ChunkTimeRef	Integer (8 bytes)			Read only. Reset during I2C reset command
Time Counter	8 ns precision, for the duration of a session. To be filled by Grabber	ChunkTimeCounter	Integer (8 bytes)			Read Only, free running counter

Table 8 – ChunkDataControl Gen<i>Cam parameters

The following table describes structure information within the raw data frame, presented byte by byte:

Field	Byte/Char	Definition	Field Generated by
XDLX	4 Char	XDlinx Labs Identity	Satellite Bus (Xdlinx Labs)
SAT_ID	4 Char	Satellite ID	Satellite Bus (Xdlinx Labs)
Orbit Number	2 Bytes	Present Orbit No	Satellite Bus (Xdlinx Labs)
Task ID	4 Bytes	Unique Job/Task ID Generated by Ground	Satellite Bus (Xdlinx Labs)
Image Start Time	4 Bytes	Imaging Start time (Seconds level Accuracy)	Satellite Bus (Xdlinx Labs)
Imaging Duration	1 Bytes	Imaging duration (Seconds level Accuracy; <32 sec[TBD])	Satellite Bus (Xdlinx Labs)
Config File + TDI File	4+4 bits= 1 Byte	Camera Configuration ID (max 16 ID); TDI Configurations (max 16 ID)	Satellite Bus (Xdlinx Labs)
Latitude	4 Bytes	Image Frame central Latitude (single precision Floating point)	Satellite Bus (Xdlinx Labs)
Longitude	4 Bytes	Image Frame central Longitude (single precision Floating point)	Satellite Bus (Xdlinx Labs)
PPS Ref	4 bytes	Seconds level precision for the duration of mission life	Camera Card (Kaya)
Time Ref	8 bytes	8 ns level precision for the duration of mission life	Camera Card (Kaya)
Time Counter	8 bytes	8 ns precision, for the duration of a session	Camera Card (Kaya)
Band used	1 byte	Number of used bands	Camera Card (Kaya)
1 st Band active	1 byte	Band number (0-6) of 1 st used band (0xFF inactive)	Camera Card (Kaya)
2 nd Band active	1 byte	Band number (0-6) of 2 nd used band (0xFF inactive)	Camera Card (Kaya)
3 rd Band active	1 byte	Band number (0-6) of 3 rd used band (0xFF inactive)	Camera Card (Kaya)
4 th Band active	1 byte	Band number (0-6) of 4 th used band (0xFF inactive)	Camera Card (Kaya)
5 th Band active	1 byte	Band number (0-6) of 5 th used band (0xFF inactive)	Camera Card (Kaya)
6 th Band active	1 byte	Band number (0-6) of 6 th used band (0xFF inactive)	Camera Card (Kaya)
7 th Band active	1 byte	Band number (0-6) of 7 th used band (0xFF inactive)	Camera Card (Kaya)
Total	56 bytes		

Table 9 – Frame MetaData structure

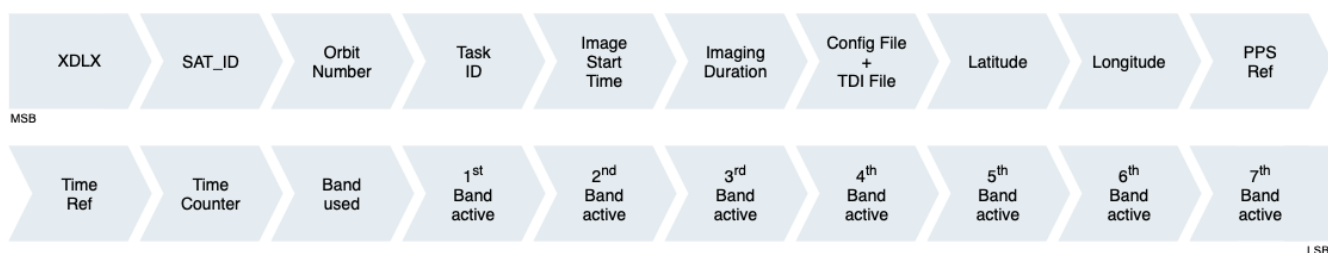


Figure 8 – Raw data frame structure

6.8 Output Buffer

The stream data is captured from the camera and stored in buffer memory. The buffer consists of enabled bands and an additional metadata line if this feature is enabled. The buffered data is then processed and transmitted for further analysis or storage.

6.8.1 Visual buffer representation

Full-frame representation when the Band1, Band3, and Band6 enabled, shown in the following figure:

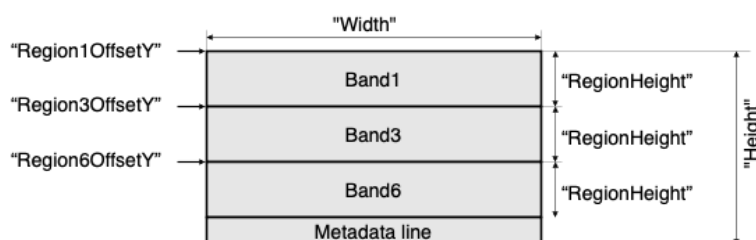


Figure 9 – Partial band selection in full frame representation

Full frame representation when all 7 bands are enabled, shown in the following figure:

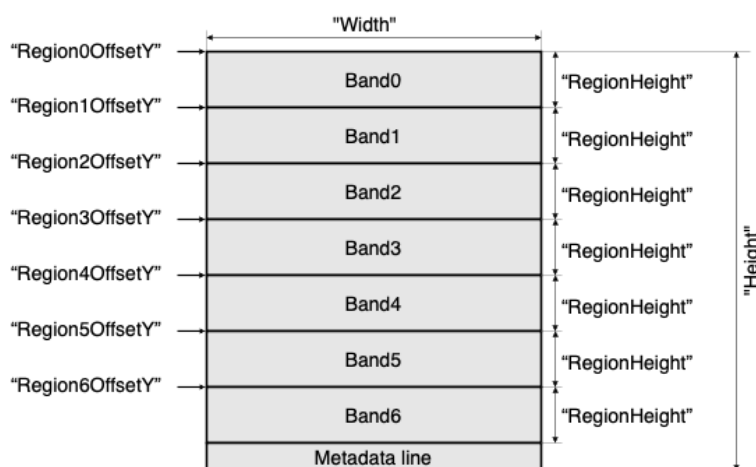


Figure 10 – Full frame representation with all bands enabled

6.8.2 Buffer size calculations

The following equations give examples of the buffer size calculation.

$$< \text{buffer_size} > = \frac{\text{“Width”} \times \text{“Height”} \times 10\text{bits}}{8}$$

$$< \text{band_size} > = \frac{\text{“Width”} \times \text{“RegionHeight”} \times 10\text{bits}}{8}$$

$$< \text{band_offset} > = < \text{band_size} > \times < \text{band_index} >$$

$$< \text{frame_metadata_line_offset} > = < \text{band_size} > \times < \text{band_count} >$$

6.9 I2C bus interface

The I2C interface is compatible to perform two write only commands, a SET command and RESET command. The I2C interface operates as a standard I2C interface with below parameters:

1. Maximum frequency 100 KHz
2. Device Address 0xA0 (DEVAD)

The transfer from master to host will be as below for different commands:

1. For RESET command host should send the below on I2C bus:
 - a. START
 - b. DEVAD + R/W=0
 - c. 0x00
 - d. 0x00
 - e. STOP
2. For SET command host should send the below on I2C bus :
 - a. START
 - b. DEVAD + R/W=0
 - c. 0x04
 - d. Send 4 bytes with the value of PPS timestamp
 - e. STOP

REFERENCES

Supported vision standards:



Vision Point documentation:



TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Instruments support service:

- Create a support request on the web: support.kayainstruments.com
- Our knowledge base is available on: kb.kayainstruments.com

Visit us at kayainstruments.com for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model
*In the Vision Point app, use menu option Help > Collect diagnostic info.		

KAYA Instruments

Please feel free to contact our sales team for pricing, availability, documentation or customization through our e-mail or phone, we will be happy to provide assistance and consultation.

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